# Find Kth Largest Value In Bst

AlgoExpert · Medium · Binary Search Trees

# The problem

- Reverse in-order traversal, stopping after k nodes.

# Intuition

- A reverse in-order traversal visits values from largest to smallest, so stop at the kth.

# Approach

- 1. Traverse right, node, left
- 2. Count visited nodes
- 3. Stop at the kth and return its value

# Java solution

```java
public class FindKthLargestValueInBst {
    private static int seen, answer;

    public static synchronized int findKthLargestValueInBst(BST tree, int k) {
        seen = 0; answer = -1;
        walk(tree, k);
        return answer;
    }

    private static void walk(BST n, int k) {
        if (n == null || seen >= k) return;
        walk(n.right, k);
        if (seen < k) { seen++; if (seen == k) { answer = n.value; return; } walk(n.left, k); }
    }
}
```

# Complexity

- Time: O(h + k)
- Space: O(h)

# Recap

- A reverse in-order traversal visits values from largest to smallest, so stop at the kth.
- Complexity: O(h + k)